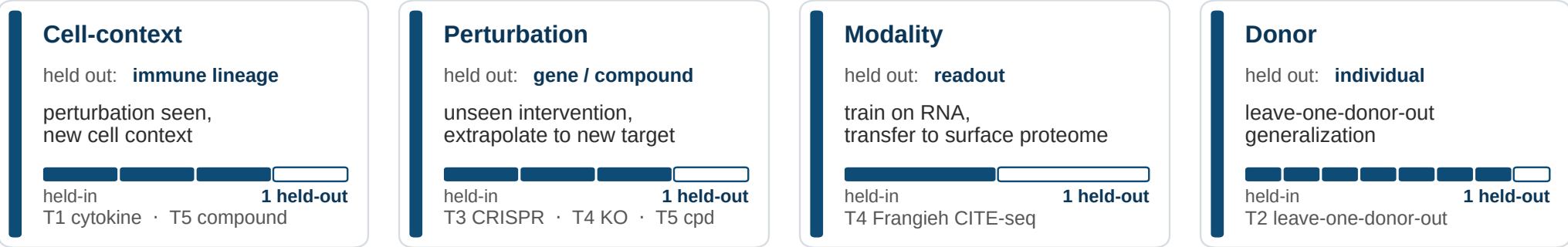
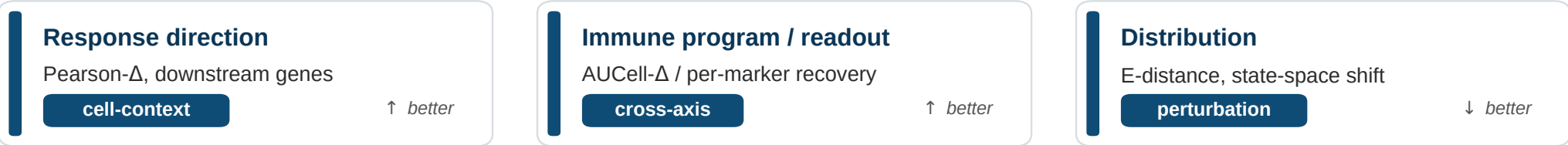


a Four immune generalization axes



b Three immune-aware metric axes



c Immune perturbation datasets

Task / source	Immune system	Perturbation	Modality	Axes tested	Methods n / 14
T1 Kang GSE96583	PBMC 8 lineages · 8 donors	IFN- β cytokine (seen)	RNA	cell-contextdonor	●●○○○○○○○○○○ 2
T2 Soskic 2022	CD4 T cells 2 lineages · 106 donors	TCR/CD28 activation (0h → 16h)	RNA	donor	●●●●○○○○○○○○ 5
T3 5× primary-T CRISPR	primary human T pan-T pool	KO / CRISPRi/a (unseen genes)	RNA	perturbation	●●●●●●○○○○○○ 9
T4 Frangieh	melanoma + TIL co-culture	≈ 248 gene KO (unseen)	RNA + 20-marker CITE	perturbationmodality	●●●●●●○○○○○○ 8
T5 OP3 GSE279945	PBMC 6 lineages · 3 donors	small-molecule compounds	RNA	cell-contextperturbation	●●●●○○○○○○○○ 6

● method run ○ not run

Methods dots = conditioned/learned methods executed on the task, of 14 distinct learned models benchmark-wide; the universal floor {cell-mean · linear-PCA} and the context comparators (control, donor-shift) run on every task and are not counted in the rail. Each model is admitted under an applicability gate (native / adapted / simple / n.a.). Soskic is shown for the donor LODO axis; temporal saturation and additional CD4-state / in-vivo extensions remain outside this benchmark panel.